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A Structured L_1 Penalization Framework for Stable Sparse Power Modeling in Water Reuse Systems

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Abstract. Water reuse and desalination facilities are increasingly exposed to time varying energy prices and operational variability, motivating digital twin models that can provide reliable, interpretable, and computationally tractable predictions in near real time. This work focuses on the empirical modeling core of the Empirical Model Based Optimization (EMBO) digital twin: adaptive L_1 -regularized regression for power forecasting from reverse osmosis (RO) flowrate setpoints. We construct a cubic polynomial feature library in permeate, concentrate, and recirculation flowrates and identify sparse regression models for power prediction. A key challenge arises during early operation or under limited data availability, where standard cross validation rules for selecting the regularization parameter (e.g., $\lambda_{\min}$ or λ_{1SE}) can lead to unstable sparsity patterns or degenerate models with all coefficients shrunk to zero. To address this, we propose a structured, λ selection strategy that explicitly controls model complexity while preventing degeneracy. The resulting model remains compact and interpretable, supports hourly retraining on rolling data windows, and provides robust power forecasts suitable for downstream decision support and optimization.

Keywords. reverse osmosis, water reuse systems, adaptive L_1 regularization.

1. Introduction

Advanced water reuse and desalination systems are increasingly operated under dynamic conditions driven by variable influent characteristics, stringent performance requirements, and fluctuations in electricity prices [1–3]. In particular, reverse osmosis (RO) typically dominates plant energy consumption, and operational degrees of freedom such as permeate, concentrate, and recirculation flowrates exert direct influence on aggregate power demand. Accurate and fast prediction of power consumption as a func-

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tion of these controlled sensor variables is therefore a critical capability for energy aware operation and for tariff responsive planning [4–6].

Digital twins (DTs) provide a framework for continuously updated virtual models that enable monitoring, prediction, and decision support for physical assets [7]. A central challenge in DT development is the identification of virtual models that are both accurate and operationally tractable [2]. High-fidelity first-principles models can be difficult to calibrate and update frequently, especially under changing process conditions. Conversely, highly flexible purely data-driven models may achieve strong predictive accuracy, but often function as black-box predictors with limited interpretability and reduced transparency for operational decision-making [2, 8]. These limitations are particularly restrictive in real-time settings, where models must be retrained regularly, executed on conventional hardware, and remain reliable even when training data are limited.

Sparse empirical modeling offers a practical compromise between fidelity and tractability. By combining a physics-consistent feature library with regularized regression, sparse models can produce interpretable relationships between control variables and power demand, while limiting overfitting and enabling rapid retraining [9–11]. In this context, the selection of the least absolute shrinkage and selection operator (LASSO) regularization parameter λ is central: overly weak regularization may lead to unstable coefficient estimates, whereas overly strong regularization can collapse the model and yield zero coefficient (degenerate) models. This risk is amplified during early operation, when only a small number of training samples are available and cross-validation error estimates exhibit high variance.

This paper focuses on the empirical modeling component of the Empirical Model-Based Optimization (EMBO) digital twin [12], with an emphasis on adaptive L_1 penalization for robust power prediction from sensor-measured flowrate variables. We develop a structured λ -selection strategy that controls sparsity while preventing degeneracy, thereby ensuring that an interpretable model remains available across operating regimes. The remainder of the paper is organized as follows. Section 2 introduces the pilot system. Section 3 presents the empirical modeling formulation, including the adaptive λ -selection logic, and evaluates the impact of structured regularization on sparsity patterns and model stability. Section 4 concludes the paper and discusses implications for real time, digital twin deployment.

2. Physical Pilot and Measurements

This section describes the experimental pilot platform used to generate training data. Figure 1(a) shows the pilot system used in this study, which replicates key operational characteristics of an advanced water reuse facility. The treatment train comprises three units in series: ultrafiltration (UF), reverse osmosis (RO), and ultraviolet/advanced oxidation (UV/AOP). Because RO is the primary energy consumer and provides direct operational degrees of freedom, this work focuses on the RO subsystem (Figure 1(b)).

Three manipulated flowrate variables are considered: permeate, concentrate, and recirculation flowrates. These quantities determine internal RO flows (e.g., feed, reject, and make up streams) and therefore provide a minimal yet complete set of inputs for power prediction. The response variable is the aggregate electrical power demand of the pilot. A model free exploratory analysis indicates strong positive associations between power de-

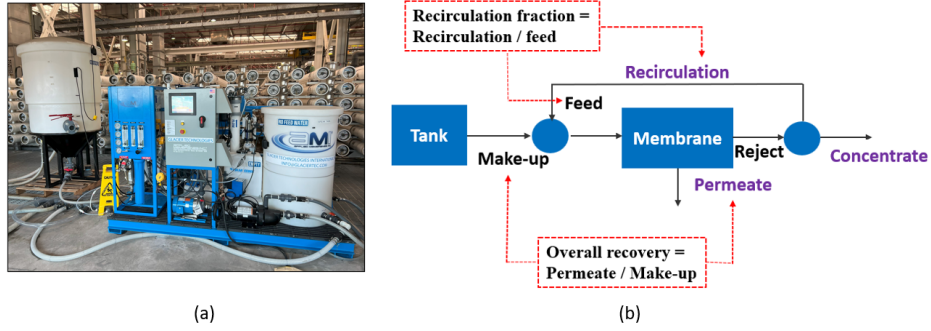


Figure 1. (a) UF-RO-UV/AOP pilot system used for data collection. (b) RO subsystem schematic highlighting the permeate, concentrate, and recirculation flowrate variables (in purple) used as inputs to the empirical model.

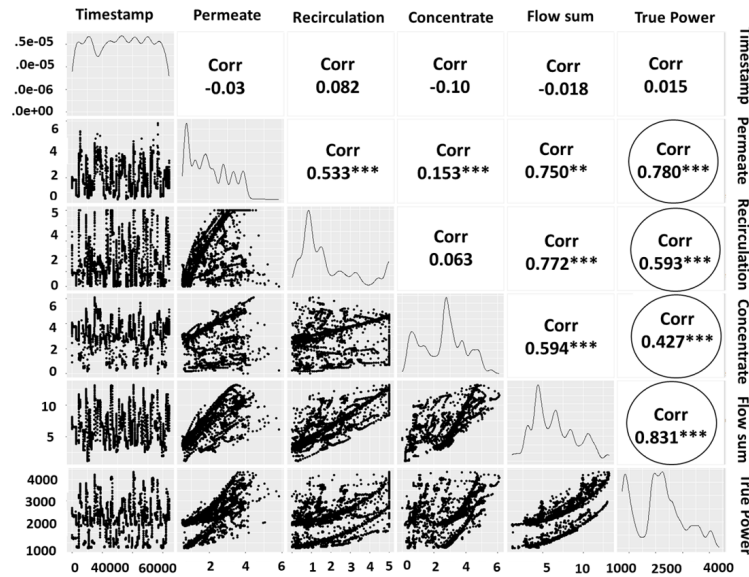


Figure 2. Pairwise scatter-plot matrix illustrating empirical relationships between aggregate power demand and flowrate variables. Pearson correlation coefficients demonstrate positive associations between power consumption and permeate, concentrate, and recirculation flowrates.

mand and each flowrate variable, with Pearson correlation coefficients of 0.780 for permeate, and 0.593 for recirculation and 0.427 for concentrate flowrates. These observed relationships motivate the polynomial model construction described next. A model-free exploratory analysis, as illustrated in figure 2, indicates strong positive associations between aggregate power demand and each flowrate variable, as well as the total flowrate (Pearson correlation ≈ 0.83). The corresponding Pearson correlation coefficients are 0.780 for permeate, 0.593 for recirculation, and 0.427 for concentrate flowrates. These observed relationships motivate the polynomial feature construction described next.

3. Digital Twin for Power Prediction & Results

A data driven digital twin was developed in Python within the EMBO framework [12]. In this short paper, we focus exclusively on the empirical modeling component that maps RO flowrate inputs to aggregate power demand. The model is trained on rolling data windows and is designed to remain reliable and interpretable across operating regimes, including early operation when only limited training data are available. Let $\mathbf{X} \in \mathbb{R}^{n \times 3}$ denote the input matrix with columns corresponding to concentrate, permeate, and recirculation flowrates and n number of observations, and let $\mathbf{Y} \in \mathbb{R}^n$ denote the measured aggregate power demand. To capture nonlinearities and cross-coupling between flowrates, we construct a polynomial feature expansion up to degree 3. With $p = 3$ input variables and maximum degree $D = 3$ (cubic polynomial), the total number of monomials excluding the constant term is $\sum_{d=1}^D \binom{p+d-1}{d} = \binom{3}{1} + \binom{4}{2} + \binom{5}{3} = 3 + 6 + 10 = 19$. Thus, the feature library contains 3 first-order terms, 6 second-order terms, and 10 third-order terms, yielding 19 candidate terms. The resulting expansion can be written as $\Phi(\mathbf{X}) = [X_1, X_2, X_3, X_1^2, X_2^2, X_3^2, X_1X_2, X_1X_3, X_2X_3, \dots]$.

Because this feature library can overfit when n is small, we estimate regression coefficients via LASSO [9, 13]: $\hat{\beta} = \arg \min_{\beta} \{ \|\mathbf{Y} - \Phi(\mathbf{X})\beta\|_2^2 + \lambda \|\beta\|_1 \}$, where $\lambda > 0$ controls the sparsity accuracy trade off. Prior to model training, both the input features and the response variable (power) are standardized to zero mean and unit variance. Consequently, the regularization parameter λ is applied in the normalized feature space. All cross-validation errors reported in this study, including the mean squared error (MSE) values in Table 1, are computed in the standardized response space and are therefore dimensionless. Larger λ promotes stronger shrinkage and fewer active terms, improving interpretability but potentially degrading predictive performance if overly aggressive.

Model training uses 10-fold cross validation over a logarithmically spaced grid $\lambda \in [10^{-4}, 10^2]$. Let $\lambda_{\min}$ denote the value that minimizes the cross validated mean squared error (CV MSE), and let λ_{1SE} denote the largest λ whose CV MSE lies within one standard error of the minimum. While λ_{1SE} is often preferred for parsimony, in early operation the cross validation error bands can be wide, and λ_{1SE} may become overly conservative, yielding very sparse or degenerate models (all coefficients shrunk to zero). Such collapse is unacceptable in a real time digital twin because it removes the predictive mapping needed for decision support. Because the polynomial feature library contains 19 candidate terms, we impose an empirical sparsity budget to maintain interpretability and robustness, particularly under limited-data conditions.

To ensure that a non degenerate and interpretable model is available throughout deployment, a structured selection rule is applied that explicitly controls model size and includes fail safe fallbacks:

1. **Primary rule:** select $\lambda_{\min}$ if the number of non-zero coefficients at $\lambda_{\min}$ is ≤ 8 (sparsity budget).
2. **Secondary rule:** otherwise scan from $\lambda_{\min}$ toward λ_{1SE} and select the first λ yielding ≤ 7 non-zero coefficients.
3. **Fallback rule:** if no such λ exists, set $\lambda_{\text{opt}} = \lambda_{1SE}$.
4. **Safeguard 1:** if $\lambda_{\text{opt}} = \lambda_{1SE}$ yields a degenerate model, revert to $\lambda_{\min}$.
5. **Safeguard 2:** if $\lambda_{\min}$ also yields zero coefficients, fit a minimal linear model via forward stepwise selection on the original (non-polynomial) inputs.

The thresholds of 8 and 7 active coefficients are heuristic sparsity budgets chosen to retain a compact subset of dominant terms (approximately fewer than half of the candidate features), rather than strict constraints. If no value of λ in $[\lambda_{\min}, \lambda_{1SE}]$ satisfies the sparsity target, the procedure defaults to λ_{1SE} .

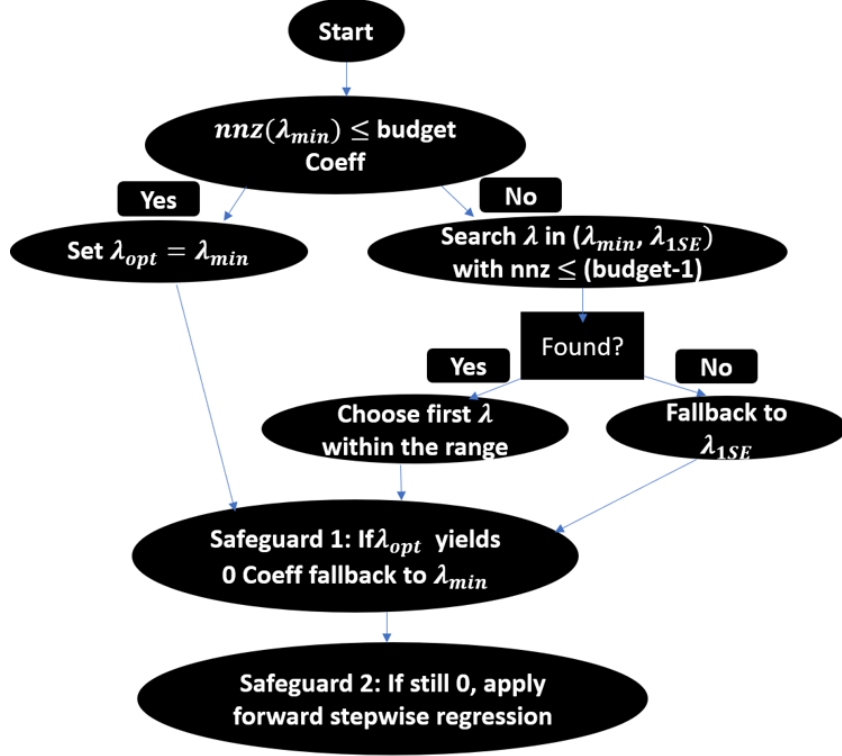


Figure 3. Structured selection and safeguard logic for adaptive L_1 regularization in the EMBO empirical model. Here nnz stands for number of nonzero coefficients.

Figure 3 summarizes this decision logic. Figure 4 and Table 1 illustrate the motivation: with sufficient data (e.g., after twenty four hours), cross validation uncertainty is reduced and conventional rules yield stable sparse models, whereas with only two hours of data, structured selection avoids premature collapse while maintaining sparsity.

Table 1. Representative cross validation outcomes illustrating structured λ selection under limited (2 hours) and expanded (24 hours) training data. MSE values are reported in standardized (dimensionless) units.

Case	$\lambda_{\min}$	λ_{1SE}	λ_{opt}	MSE $\lambda_{\min}$	MSE λ_{opt}	Nonzeros
24 hours	0.000099	0.025119	0.025119	0.148914	0.159440	11
2 hours	0.0015849	0.0794328	0.019953	0.59325	0.685191	7

Next, for any candidate flowrate input matrix $\mathbf{X}_{\text{cand}} \in \mathbb{R}^{n \times 3}$ (n is number of observations), power is predicted as $\hat{\mathbf{Y}} = \Phi(\mathbf{X}_{\text{cand}})\hat{\boldsymbol{\beta}}$. The sparse structure of $\hat{\boldsymbol{\beta}}$ provides interpretability by identifying the dominant flowrate terms and interactions that drive power

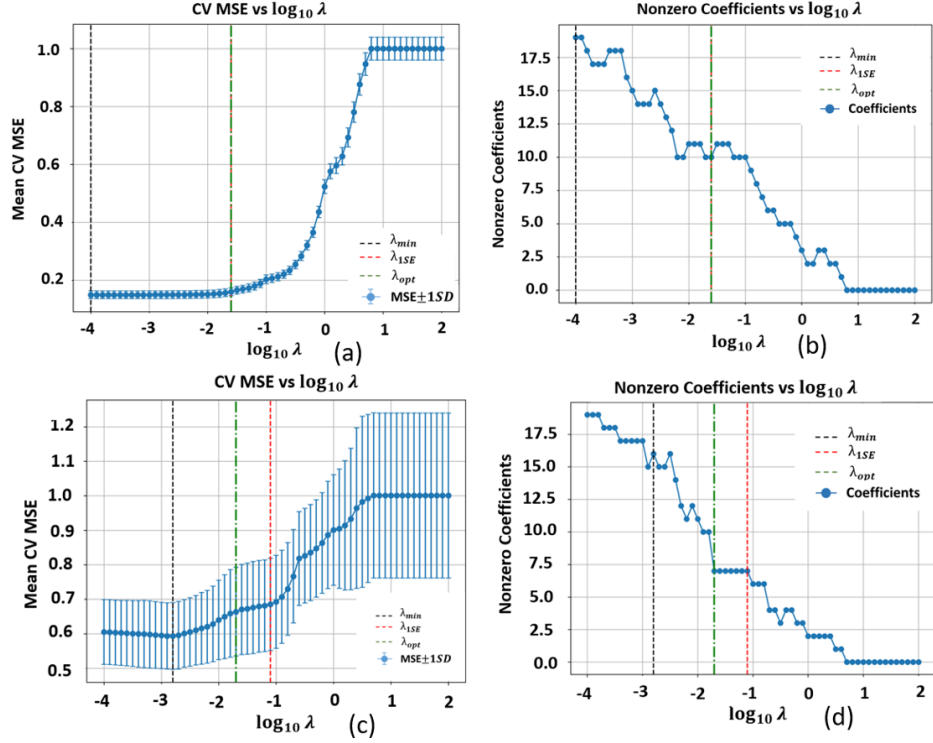


Figure 4. Cross-validation performance and sparsity behavior under different regularization levels: (a) cross-validated mean squared error (MSE) versus λ (log scale) after 24 hours of operation; (b) number of nonzero coefficients versus λ (log scale) after 24 hours of operation; (c) cross-validated MSE versus λ (log scale) after 2 hours of operation; (d) number of nonzero coefficients versus λ (log scale) after 2 hours of operation.

demand. Figure 5 (a) reports feature importance for a representative model identified after twenty four hours, where 11 of the 19 candidate terms remain active (approximately 42.1% sparsity), and first order flowrate terms dominate the learned mapping. Figure 5(b) demonstrates close agreement between the predicted power (orange) and the experimentally measured power (green), indicating good predictive fidelity of the empirical model.

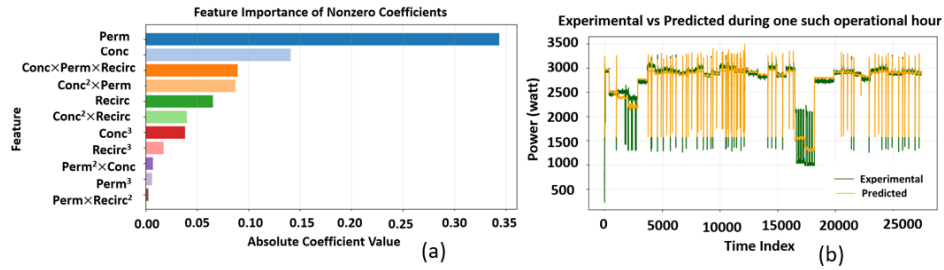


Figure 5. (a) Feature importance derived from the sparse model after 24 hours of operation where conc, perm and recirc stands for concentrate, permeate and recirculation flowrates respectively, (b) Comparison of predicted power (in orange) and experimental power (in green) during same period of operation.

4. Conclusion and Future Work

This work focuses exclusively on the empirical modeling component of a digital twin framework for water reuse systems, demonstrating that adaptive L_1 -regularized regression can provide reliable and interpretable power predictions from RO flowrate inputs under limited and evolving data conditions. By introducing a structured and fail safe regularization selection strategy, the proposed approach mitigates degeneracy and over sparsification during early operation, while preserving model stability and interpretability as training data accumulate. The results highlight the suitability of adaptive sparse coefficients as a computationally efficient foundation for real time digital twin applications operating on conventional hardware. Future work will extend this empirical modeling framework toward a digital twin, in which the learned power model is embedded within a receding horizon optimization loop that is retrained hourly on rolling data windows and used to generate flowrate setpoints that are fed back to the physical pilot for continuous real time adaptation. Future research will also investigate hybrid digital twin frameworks that combine empirical models with physics-based simulators to broaden applicability across operating regimes.

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Data Availability Statement

Data cannot be made publicly available; readers should contact the corresponding author for further details.

Conflict of Interest

The authors declare that they have no conflict of interest.

References

- [1] Pinhas Glueckstern and Deborah Anahory. Design optimization of large SWRO plants utilizing time-of-use (TOU) energy prices. *Desalination and Water Treatment*, 31(1-3):172–177, 2011.
- [2] Jan Kloppenborg Møller, Goran Goranović, Per Brath, and Henrik Madsen. A data-driven digital twin for water ultrafiltration. *Communications Engineering*, 1(1):23, 2022.

- [3] Vladimir Z Gjorgievski, Bodan Velkovski, Blagoja Markovski, Snezana Cundeva, and Natasa Markovska. Energy community demand-side flexibility: Energy storage and electricity tariff synergies. *Energy*, 313:134078, 2024.
- [4] Mohammad Amin Soleimanzade, Amit Kumar, and Mohtada Sadrzadeh. Novel data-driven energy management of a hybrid photovoltaic-reverse osmosis desalination system using deep reinforcement learning. *Applied Energy*, 317:119184, 2022.
- [5] Yang Liu, Jun Feng, Jiamin Lu, and Siyuan Zhou. A review of digital twin capabilities, technologies, and applications based on the maturity model. *Advanced Engineering Informatics*, 62:102592, 2024.
- [6] Faegheh Moazeni and Javad Khazaei. Optimal design and operation of an islanded water-energy network including a combined electrodialysis-reverse osmosis desalination unit. *Renewable Energy*, 167:395–408, 2021.
- [7] Prabhakar Sharma, Bhaskor Jyoti Bora, B Deepanraj, and T Jarin. Overview of digital twins in renewable energy. In *2024 1st International Conference on Trends in Engineering Systems and Technologies (ICTEST)*, pages 1–6. IEEE, 2024.
- [8] Yanwen Shi, Jie Xu, and Weijia Du. Discussion on the new operation management mode of hydraulic engineering based on the digital twin technique. In *Journal of Physics: Conference Series*, volume 1168, page 022044. IOP Publishing, 2019.
- [9] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1):267–288, 1996.
- [10] Robert Tibshirani. Regression shrinkage and selection via the lasso: a retrospective. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 73(3): 273–282, 2011.
- [11] Valeria Fonti and Eduard Belitser. Feature selection using lasso. *VU Amsterdam research paper in business analytics*, 30:1–25, 2017.
- [12] Subrata Mukherjee. EMBO framework: Empirical model-based optimization for digital-physical twin control, 2025. URL https://github.com/submukherjee/EMBO_framework. Accessed: January 2026.
- [13] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor. Linear model selection and regularization. In *An Introduction to Statistical Learning: with Applications in Python*, pages 229–288. Springer, 2023.